BCMS residual sensitivity: background difference and forecast horizon

High + extreme deviation groups · frozen-output removal

Candidate-background difference

(a) Chengdu · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 79.0k–82.6k	+2.175	+2.355	+3.074	+2.352	+1.287	+1.918	+2.001	+2.575	+2.859	+3.643	+2.639	+2.232
Medium n = 99.4k–102.1k	+1.405	+1.549	+2.075	+1.602	+0.752	+1.269	+1.346	+1.739	+1.945	+2.536	+1.784	+1.465
High n = 126.5k–130.1k	+0.639	+0.725	+1.051	+0.917	+0.381	+0.653	+0.645	+0.844	+0.949	+1.193	+0.839	+0.606

(b) Chengdu · outflow

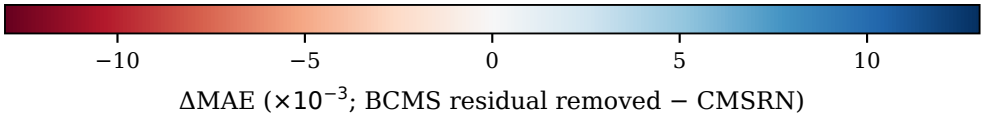
	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 85.8k–88.8k	+1.594	+1.316	+0.826	+1.089	+0.244	+0.481	+0.853	+1.419	+0.732	+2.018	+1.127	+0.483
Medium n = 111.3k–115.2k	+1.063	+0.911	+0.543	+0.617	+0.073	+0.184	+0.605	+0.891	+0.486	+1.430	+0.739	+0.256
High n = 114.9k–119.5k	+0.450	+0.386	+0.162	+0.233	-0.005	+0.069	+0.276	+0.435	+0.222	+0.725	+0.312	+0.115

(c) Shanghai · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 91.4k–95.1k	+12.350	+12.530	+12.671	+12.785	+12.213	+12.448	+12.315	+12.432	+13.002	+12.413	+12.700	+12.701
Medium n = 112.5k–115.9k	+8.895	+8.813	+9.068	+8.972	+8.626	+8.885	+8.782	+9.069	+9.342	+8.645	+9.120	+9.027
High n = 129.0k–131.0k	+6.166	+6.073	+6.212	+6.424	+6.295	+6.301	+6.151	+6.643	+6.855	+6.117	+6.562	+6.594

(d) Shanghai · outflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 90.6k–93.4k	+10.203	+9.899	+10.119	+10.149	+9.849	+9.809	+10.054	+9.990	+9.951	+9.914	+9.982	+10.146
Medium n = 110.7k–114.2k	+7.568	+7.404	+7.742	+7.645	+7.334	+7.452	+7.691	+7.601	+7.569	+7.396	+7.489	+7.561
High n = 131.0k–134.2k	+6.022	+5.793	+5.914	+5.841	+5.678	+5.771	+5.876	+5.601	+5.708	+5.541	+5.674	+5.523



ΔMAE (×10^{-3}; BCMS residual removed − CMSRN)

Values show 1000 × ΔMAE; divide by 1000 to recover original flow units.
ΔMAE = MAE(prediction − BCMS residual) − MAE(CMSRN prediction).
Frozen-output removal, not a retrained ablation or an isolated background test.
H1–H12: 5–60 min. Blue: helps; red: hurts. Counts span valid pairs across horizons.